

# Electronic Circuits

A complete syllabus-based digital textbook for transistor biasing, single-stage and multistage amplifiers, tuned and power amplifiers, feedback, sinusoidal oscillators, transistor switching, multivibrators, Schmitt trigger and UJT relaxation oscillator—supported by Malayalam notes, formulas, diagrams and solved design problems.

**COURSE CODE****3043****COURSE TITLE****Electronic Circuits****SEMESTER****3****CREDITS****4****CATEGORY****Program Core****PERIODS / WEEK****4 (L:3, T:1, P:0)****PERIODS / SEMESTER****60****MAIN LEVEL****Applying**

# 60

## Total periods

Four course outcomes, twenty-one module outcomes, fifty-eight teaching hours and two series-test periods.

**COURSE PURPOSE**

## What this course develops

### Amplifier design

Choose biasing, calculate the operating point and develop single-stage and cascaded transistor amplifiers.

Biasing, operating point കണക്കാക്കുക, single-stage, multistage transistor amplifiers രൂപകൽപ്പന ചെയ്യുക.

## Frequency selection

Use resonance and tuned circuits to amplify a selected frequency band.

Resonance, tuned circuits എന്നിവ ഉപയോഗിച്ച് ആവശ്യമായ frequency band മാത്രം amplify ചെയ്യുക.

## Power delivery

Compare amplifier classes and understand output power, efficiency, impedance matching and crossover distortion.

Power amplifier classes, output power, efficiency, impedance matching, crossover distortion എന്നിവ താരതമ്യം ചെയ്യുക.

## Stable gain

Apply negative feedback to improve gain stability, bandwidth, distortion and impedances.

Negative feedback ഉപയോഗിച്ച് gain stability, bandwidth, distortion, input/output impedance എന്നിവ മെച്ചപ്പെടുത്തുക.

## Waveform generation

Design RC, LC and crystal oscillators and explain their frequency-determining networks.

RC, LC, crystal oscillators രൂപകൽപ്പന ചെയ്ത് frequency നിർണ്ണയിക്കുന്ന network വിശദീകരിക്കുക.

## Pulse and switching circuits

Use BJTs and UJTs for switching, multivibrators, Schmitt triggering and relaxation oscillation.

BJT/UJT ഉപയോഗിച്ച് switching, multivibrator, Schmitt trigger, relaxation oscillator circuits നിർമ്മിക്കുക.

# Module and course-outcome map

The PDF allocates 58 instructional hours plus two series-test periods, giving the stated semester total of 60 periods.

| CO1 / Module 1                                                                                                                                                                                                                            |                                                          |          | 14 hours           |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|----------|--------------------|
| CO2 / Module 2                                                                                                                                                                                                                            |                                                          |          | 15 hours + Test I  |
| CO3 / Module 3                                                                                                                                                                                                                            |                                                          |          | 15 hours           |
| CO4 / Module 4                                                                                                                                                                                                                            |                                                          |          | 14 hours + Test II |
| CO                                                                                                                                                                                                                                        | Outcome                                                  | Duration | Cognitive level    |
| CO1                                                                                                                                                                                                                                       | Develop basic single-stage and multistage amplifiers.    | 14 hours | Applying           |
| CO2                                                                                                                                                                                                                                       | Develop basic tuned amplifiers and power amplifiers.     | 15 hours | Applying           |
| CO3                                                                                                                                                                                                                                       | Develop feedback amplifiers and sinusoidal oscillators.  | 15 hours | Applying           |
| CO4                                                                                                                                                                                                                                       | Use transistors to realise pulse and switching circuits. | 14 hours | Applying           |
| <b>Prerequisites:</b> AC signal representation and passive components; semiconductor theory, diodes and transistors; basic Mathematics I and II principles. All four course outcomes are strongly mapped to PO1 in the supplied syllabus. |                                                          |          |                    |

CO1 • 14 HOURS

## Module 1 — Single-Stage and Multistage Amplifiers

Biasing establishes a stable Q-point; the amplifier then converts a small AC input into a larger output without unwanted distortion.

M1.01

3 hours • Understanding

### Transistor biasing, DC load line and operating point

**Biasing** supplies the DC voltages and currents required to keep a transistor in the active region. The selected operating point or **Q-point** should permit the signal to swing in both directions without entering cutoff or saturation.

**DC load line:**  $V_{CE} = V_{CC} - I_C R_C$  | endpoints: ( $I_C=0$ ,  $V_{CE}=V_{CC}$ ) and ( $V_{CE}=0$ ,  $I_C=V_{CC}/R_C$ )

Temperature and transistor  $\beta$  variation can move the Q-point. Stabilisation reduces this movement. Fixed bias is simple but poorly stabilised; voltage-divider bias with an emitter resistor provides much better stability.

## Fixed bias

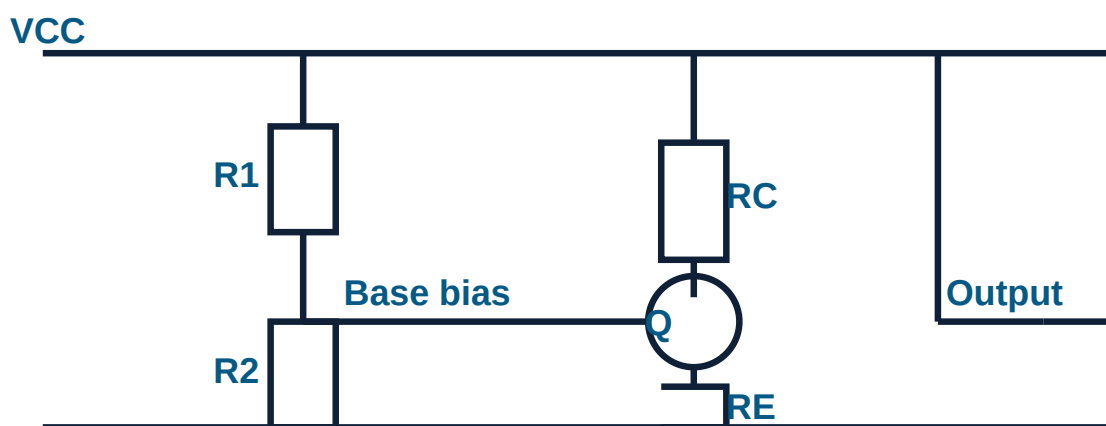
A resistor from  $V_{CC}$  to base sets base current. It uses few components but the collector current depends strongly on  $\beta$  and temperature.

## Voltage-divider bias

$R_1$  and  $R_2$  establish a nearly fixed base voltage, while  $R_E$  introduces negative DC feedback and stabilises the Q-point.

**Approximate divider design:**  $V_B = V_{CC}R_2/(R_1 + R_2)$ ,  $V_E \approx V_B - 0.7 \text{ V}$ ,  $I_E \approx V_E/R_E$ ,  $V_{CE} \approx V_{CC} - I_C R_C - I_E R_E$

**മലയാളം സഹായക്കുറിപ്പ്:** Transistor active region-ൽ സ്ഥിരമായി പ്രവർത്തിക്കാൻ DC voltage/current നൽകുന്നതാണ് biasing. Q-point cutoff-നും saturation-നും നടുവിലായി വെച്ചാൽ output clipping കുറയും. Voltage-divider bias-ൽ emitter resistor ഉള്ളതിനാൽ stability കൂടുതലാണ്.



M1.02

4 hours • Applying

## Single-stage CE amplifier with voltage-divider bias

In a common-emitter amplifier, the input is applied between base and emitter and the output is taken between collector and emitter. The stage provides voltage, current and power gain, and the output voltage is approximately  $180^\circ$  out of phase with the input.

$$r'_e \approx 25 \text{ mV}/I_E; \quad A_v \approx -(R_C \parallel R_L)/r'_e; \quad A_i \approx \beta; \quad A_p = |A_v|A_i; \quad BW = f_H - f_L$$

- ✓ Input coupling capacitor blocks source DC.
- ✓ Emitter bypass capacitor increases midband voltage gain by providing an AC ground.
- ✓ Output coupling capacitor blocks collector DC from the load.
- ✓ At low frequency, coupling and bypass capacitors reduce gain.

**മലയാളം സഹായക്കുറിപ്പ്:** CE amplifier നല്ല voltage gain നൽകുന്നു; output input-നോട്  $180^\circ$  phase reversal കാണിക്കും. Coupling capacitors DC തടയും. Emitter bypass capacitor AC gain വർദ്ധിപ്പിക്കും. Midband-ൽ gain ഏകദേശം constant ആണ്.

### Solved design example — voltage-divider Q-point

Given  $V_{CC}=12\text{ V}$ ,  $R_1=47\text{ k}\Omega$ ,  $R_2=10\text{ k}\Omega$ ,  $R_C=2.2\text{ k}\Omega$  and  $R_E=1\text{ k}\Omega$ :

- 1  $V_B=12 \times 10 / (47+10)=2.105\text{ V}$ .
- 2  $V_E \approx 2.105 - 0.7 = 1.405\text{ V}$ .
- 3  $I_E \approx 1.405 / 1\text{ k}\Omega = 1.405\text{ mA}$  and  $I_C \approx 1.405\text{ mA}$ .
- 4  $V_C = 12 - (1.405\text{ mA} \times 2.2\text{ k}\Omega) = 8.909\text{ V}$ .
- 5  $V_{CE} = 8.909 - 1.405 = 7.504\text{ V}$ . The Q-point is approximately (7.50 V, 1.41 mA).

### Solved gain example

For  $I_E=1\text{ mA}$  and effective AC collector load  $2\text{ k}\Omega$ :  $r'_e \approx 25\text{ }\Omega$ , therefore  $A_v \approx -2000/25 = -80$ . The magnitude of voltage gain is 80 and the negative sign indicates phase reversal.

M1.03

1 hour • Understanding

## Emitter follower

The emitter follower or common-collector stage takes output from the emitter. Its voltage gain is slightly less than unity, but it provides high input impedance, low output impedance and large current gain.

$$A_v \approx 1; A_i \approx \beta + 1$$

**Applications:** buffering, impedance matching, isolation between stages and driving a low-resistance load.

**മലയാളം:** Emitter follower-ന് voltage gain ഏകദേശം 1 ആണ്. High input impedance, low output impedance ഉള്ളതിനാൽ buffer, impedance matching stage ആയി ഉപയോഗിക്കുന്നു.

M1.04

3 hours • Applying

## Multistage amplifiers and coupling methods

When one stage cannot provide sufficient gain or impedance transformation, two or more stages are cascaded. The output of one stage drives the input of the next.

$A_{v,total} = A_{v1}A_{v2} \dots A_{vn}$ ; Gain(dB) =  $20 \log_{10}|A_v|$ ; total dB gain = sum of stage dB gains

| Coupling             | Strength                                             | Limitation                                                      | Typical use                                     |
|----------------------|------------------------------------------------------|-----------------------------------------------------------------|-------------------------------------------------|
| RC coupling          | Good audio-frequency response, inexpensive, compact. | Poor impedance matching and reduced gain at frequency extremes. | Voltage amplifiers.                             |
| Transformer coupling | Excellent impedance matching and power transfer.     | Bulky, costly and narrower frequency response.                  | Power-output stages.                            |
| Direct coupling      | Passes DC and very low frequencies.                  | Drift and bias interaction between stages.                      | DC amplifiers, instrumentation and IC circuits. |

**മലയാളം:** Multistage amplifier-ൽ stages cascade ചെയ്യുന്നു. Overall gain = ഓരോ stage gain-ന്റെയും product. RC coupling audio voltage amplification-ക്ക്; transformer coupling impedance matching-ക്ക്; direct coupling DC/low-frequency signals-ക്ക് അനുയോജ്യം.

### Solved overall-gain example

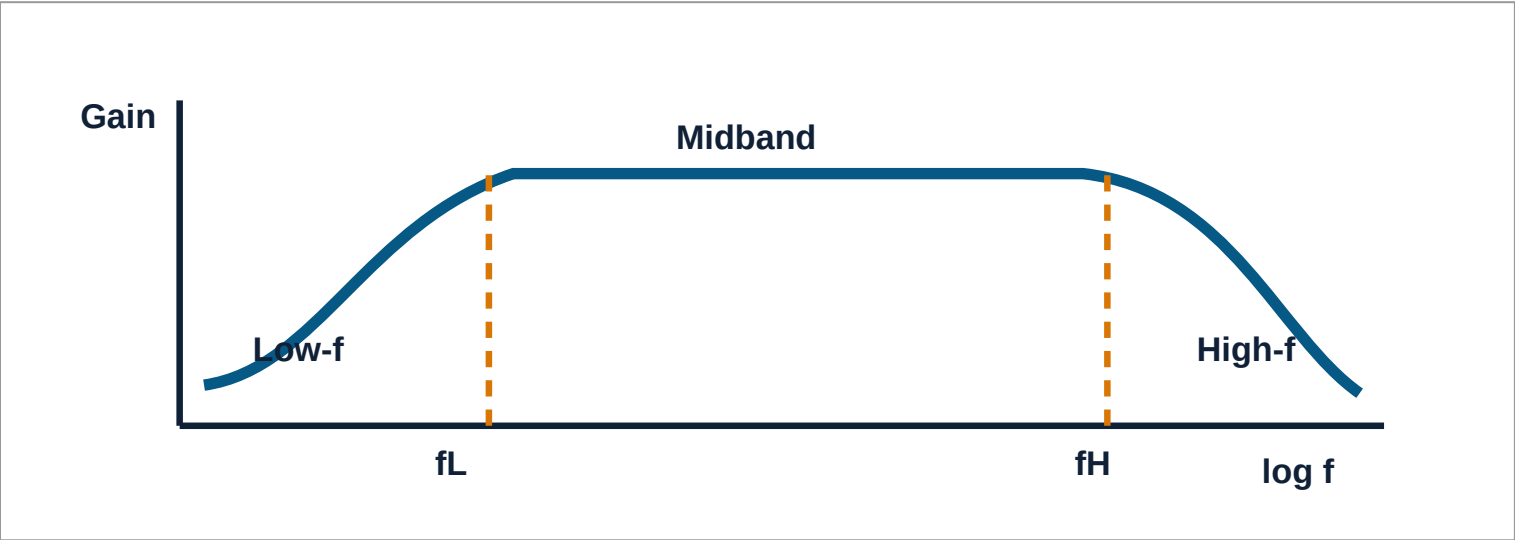
For  $A_{v1}$ =20 and  $A_{v2}$ =30, total gain=600. Gain in dB= $20 \log_{10}(600)$ =55.56 dB.

M1.05–M1.06

3 hours • Understanding / Remembering

## Frequency response and applications of multistage amplifiers

The response has three regions. In the low-frequency region, coupling and bypass capacitor reactance is significant. In the midband region, gain is nearly constant. At high frequency, transistor junction and stray capacitances cause gain reduction. Cascading stages generally narrows the overall bandwidth.



**Applications:** audio preamplifiers, microphone amplifiers, instrumentation channels, communication receivers, sensor interfaces and driver stages.

**മലയാളം:** Frequency response-ൽ low, midband, high എന്നീ മൂന്ന് മേഖലകൾ ഉണ്ട്. Bandwidth =  $f_H - f_L$ . Stages കൂടുതലാകുമ്പോൾ overall bandwidth സാധാരണ കുറയും.

**Module 1 exam pattern:** Draw the bias circuit, locate the Q-point, write the CE gain expressions, compare coupling schemes and solve at least one Q-point or cascaded-gain problem.

CO2 • 15 HOURS + SERIES TEST I

## Module 2 — Tuned Amplifiers and Power Amplifiers

Tuned amplifiers select a narrow frequency band; power amplifiers deliver substantial AC power to a load efficiently.

M2.01–M2.02

6 hours • Applying / Understanding

### Resonance, single-tuned and double-tuned amplifiers

At resonance, inductive and capacitive reactances cancel. A series resonant circuit has minimum impedance and maximum current. A parallel resonant circuit has maximum impedance and behaves as a selective load for a tuned amplifier.

$$f_r=1/(2\pi\sqrt{LC}); \quad Q=f_r/BW; \quad BW=f_r/Q; \quad \text{series } Q=\omega_0L/R=1/(\omega_0CR)$$

#### Single-tuned amplifier

One parallel LC circuit forms the collector load. Gain peaks near resonance. It is simple and selective, but the gain–bandwidth compromise limits performance.

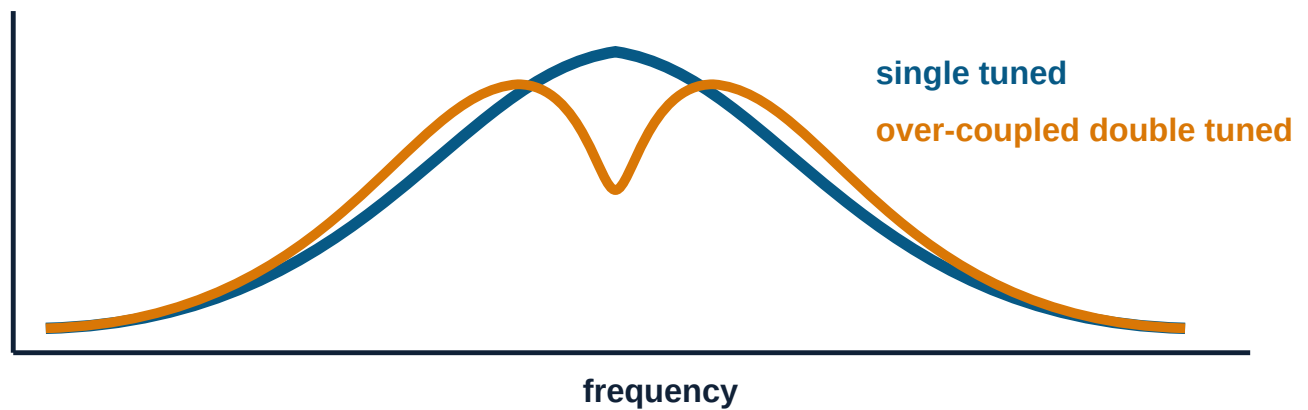
#### Double-tuned amplifier

Two magnetically coupled tuned circuits provide improved gain and adjustable bandwidth. Under-coupling gives one narrow peak, critical coupling gives a broad flat peak, and over-coupling gives two peaks.

**മലയാളം:** Resonance-ൽ L, C reactance-കൾ cancel ചെയ്യും. Single tuned amplifier-ൽ ഒരു LC tank; double tuned-ൽ coupled ആയ രണ്ട് tanks. Q കൂടുമ്പോൾ selectivity കൂടുകയും bandwidth കുറയുകയും ചെയ്യും.

**Solved tuned-circuit example**

For L=100 μH and C=100 pF:  $f_r=1/[2\pi\sqrt{(100\times10^{-6}\times100\times10^{-12})}]=1.59\text{ MHz}$ . If Q=50, BW=1.59 MHz/50=31.8 kHz.



## M2.03–M2.05

9 hours • Applying / Understanding / Remembering

## Power amplifiers, classes and push-pull operation

A voltage amplifier mainly raises signal voltage; a power amplifier transfers maximum useful AC power to the load. Power stages use large-signal operation, heat dissipation and impedance matching.

| Class | Conduction angle               | Efficiency / feature                                                                                                              | Main use                            |
|-------|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| A     | 360°                           | Best linearity; low efficiency. Maximum theoretical efficiency: 25% with resistive collector load, 50% with transformer coupling. | Low-distortion small-power stages.  |
| B     | 180°                           | Maximum theoretical push-pull efficiency 78.5%; crossover distortion near zero crossing.                                          | Audio power output.                 |
| AB    | More than 180°, less than 360° | Reduces crossover distortion; efficiency between Class A and B.                                                                   | Most practical audio output stages. |
| C     | Less than 180°                 | High efficiency but severe waveform distortion; requires tuned load.                                                              | RF power amplification.             |

## Single-ended versus double-ended

A single-ended stage uses one active device for the complete output. A double-ended or push-pull stage uses two devices on alternate half cycles. Push-pull cancels even harmonics, increases output power and avoids DC magnetisation in a centre-tapped output transformer, but requires matched devices and phase splitting.

$$\text{Class-B push-pull: } P_{o,\max} = V_{CC}^2 / (2R_L); \quad P_{dc} = 2V_{CC}I_m / \pi; \quad \eta_{\max} = \pi/4 = 78.5\%$$

**മലയാളം:** Voltage amplifier signal voltage വർദ്ധിപ്പിക്കുന്നു; power amplifier load-ലേക്ക് power നൽകുന്നു. Class A linear പക്ഷേ efficiency കുറവ്. Class B push-pull efficiency കൂടുതലെങ്കിലും crossover distortion ഉണ്ടാകും. Class AB അതിനെ കുറയ്ക്കുന്നു. Class C RF tuned circuits-ൽ ഉപയോഗിക്കുന്നു.



### Solved Class-B example

For  $V_{CC}=12\text{ V}$  and  $R_L=8\ \Omega$ , maximum AC output power is  $12^2/(2 \times 8)=9\text{ W}$ . Peak load current is  $12/8=1.5\text{ A}$ . DC input power is  $2 \times 12 \times 1.5/\pi=11.46\text{ W}$ . Efficiency= $9/11.46=78.5\%$ .

**Common mistake:** Do not state that Class B has zero distortion. Ideal devices produce a dead zone around zero due to base-emitter threshold; Class AB bias is used to reduce this crossover distortion.

**Series Test I focus:** resonance and Q, single versus double tuning, power-amplifier classes, Class-B push-pull working, efficiency expressions, advantages, disadvantages and applications.

CO3 • 15 HOURS

## Module 3 — Feedback Amplifiers and Sinusoidal Oscillators

Feedback controls amplifier behaviour; positive regenerative feedback is used with a frequency-selective network to sustain oscillation.

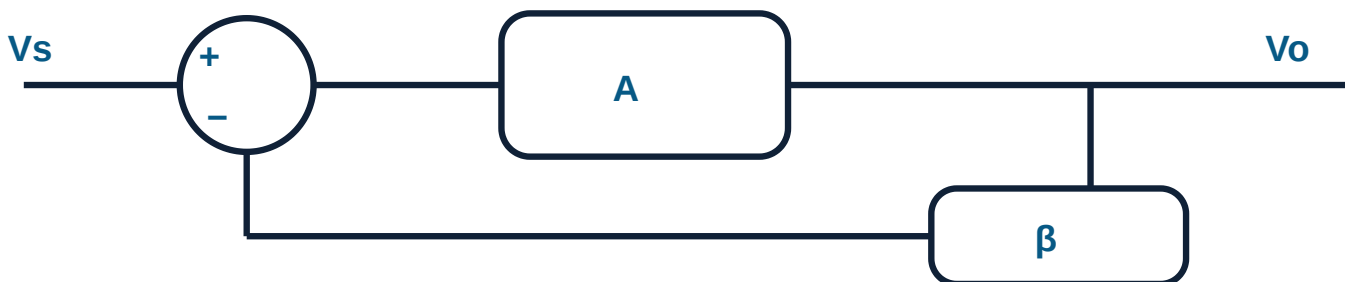
M3.01–M3.03

7 hours • Understanding

### Positive and negative feedback

Feedback returns a fraction  $\beta$  of the output to the input. If the returned signal aids the input, feedback is positive. If it opposes the input, feedback is negative.

Negative feedback gain:  $A_f=A/(1+A\beta)$  | Positive feedback gain:  $A_f=A/(1-A\beta)$



### Benefits of negative feedback

✓ Stabilises closed-loop gain against device and temperature variation.

✓ Reduces nonlinear distortion and internally generated noise approximately by the desensitivity factor  $1+A\beta$ .

- ✓ Increases bandwidth.  
Course 3043
- ✓ Allows input and output impedance to be modified by the connection topology.

| Topology       | Input mixing | Output sampling | $Z_{in}$  | $Z_{out}$ |
|----------------|--------------|-----------------|-----------|-----------|
| Voltage-series | Series       | Voltage         | Increases | Decreases |
| Voltage-shunt  | Shunt        | Voltage         | Decreases | Decreases |
| Current-series | Series       | Current         | Increases | Increases |
| Current-shunt  | Shunt        | Current         | Decreases | Increases |

The syllabus specifically includes typical current-series and voltage-series transistor feedback circuits.

**മലയാളം:** Output-ന്റെ ഒരു ഭാഗം input-ലേക്ക് തിരികെ നൽകുന്നതാണ് feedback. Negative feedback gain കുറയ്ക്കുമ്പോഴും stability, bandwidth എന്നിവ വർദ്ധിപ്പിക്കുകയും distortion കുറയ്ക്കുകയും ചെയ്യും. Series mixing input impedance കൂട്ടും; shunt mixing കുറക്കും.

**Solved feedback example**

Open-loop gain  $A=100$  and  $\beta=0.04$ . Loop gain  $A\beta=4$ . Closed-loop gain  $A_f=100/(1+4)=20$ . The desensitivity factor is 5, so parameter variation and distortion are reduced by approximately five times.

## Oscillation principle and Barkhausen criterion

An oscillator converts DC supply power into an AC waveform without an external periodic input. A frequency-selective network returns a correctly phased signal to the amplifier.

**Barkhausen criterion:**  $|A\beta|=1$  and total loop phase shift =  $0^\circ$  or  $360^\circ$  at the oscillation frequency.

| Oscillator     | Frequency expression                                      | Key point                                                              | Application                                                 |
|----------------|-----------------------------------------------------------|------------------------------------------------------------------------|-------------------------------------------------------------|
| RC phase-shift | $f=1/(2\pi RC\sqrt{6})$ for three equal RC sections       | Amplifier gives $180^\circ$ and RC network gives another $180^\circ$ . | Audio-frequency sine generation.                            |
| Wien bridge    | $f=1/(2\pi RC)$ for equal R and C                         | Low distortion; amplifier gain condition is approximately 3.           | Audio test oscillator.                                      |
| Hartley        | $f=1/[2\pi\sqrt{C(L_1+L_2+2M)}]$                          | Inductive divider supplies feedback.                                   | RF signal generation.                                       |
| Colpitts       | $f=1/(2\pi\sqrt{LC_{eq}})$ ,<br>$C_{eq}=C_1C_2/(C_1+C_2)$ | Capacitive divider supplies feedback.                                  | Stable RF oscillator.                                       |
| Crystal        | Set mainly by crystal mechanical resonance                | Piezoelectric crystal has extremely high Q and excellent stability.    | Clocks, communication transmitters and frequency standards. |

**മലയാളം:** Oscillator external AC input ഇല്ലാതെ DC-യിൽ നിന്ന് periodic AC waveform ഉണ്ടാക്കുന്നു. Barkhausen condition: loop gain magnitude 1, phase shift 0°/360°. RC oscillators audio frequency-ക്ക്, LC oscillators RF-ക്ക്, crystal oscillator വളരെ stable frequency-ക്ക് ഉപയോഗിക്കുന്നു.

**Design example — RC phase-shift oscillator at 1 kHz**

Choose C=0.01 μF=10 nF.  $R=1/(2\pi fC\sqrt{6})=1/[2\pi\times1000\times10\text{ nF}\times\sqrt{6}]=6.50\text{ k}\Omega$ . Use a nearby standard value such as 6.49 kΩ or 6.8 kΩ and verify the practical frequency.

**Design example — Wien bridge oscillator at 1 kHz**

For C=0.01 μF,  $R=1/(2\pi fC)=15.9\text{ k}\Omega$ . Use R≈15.9 kΩ in both arms and set the amplifier closed-loop gain near 3.

**Applications of sinusoidal oscillators**

Signal generators, radio transmitters and receivers, local oscillators, clocks, function generators, audio testing, carrier generation and measurement instruments.

**Module 3 exam pattern:** Derive A<sub>f</sub>, explain the four feedback topologies, list negative-feedback effects, state Barkhausen criterion and design at least one RC or LC oscillator.

**CO4 • 14 HOURS + SERIES TEST II**

**Module 4 — Pulse and Switching Circuits**

Transistors can be driven between cutoff and saturation to act as electronic switches and to generate rectangular, pulse and relaxation waveforms.

M4.01

1 hour • Understanding

**Transistor as a switch**

| State      | Condition                                                                 | Equivalent    |
|------------|---------------------------------------------------------------------------|---------------|
| Cutoff     | Base current approximately zero; $I_C\approx0$ ; $V_{CE}\approx V_{CC}$ . | Open switch   |
| Saturation | Sufficient base drive; $V_{CE(sat)}$ small, typically about 0.2 V.        | Closed switch |

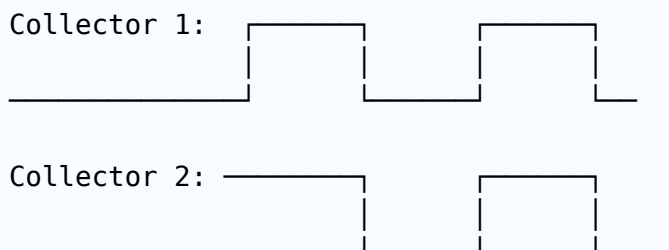
Forced base design:  $I_B \geq I_C/\beta_{forced}$ ;  $R_B=(V_{drive}-V_{BE})/I_B$

**മലയാളം:** Cutoff-ൽ transistor OFF/open switch; saturation-ൽ ON/closed switch. Reliable saturation നേടാൻ ആവശ്യമായ base current നൽകണം.

## BJT astable multivibrator

An astable multivibrator has no stable state. Two transistor stages switch alternately through cross-coupled capacitors, producing complementary rectangular collector waveforms.

$$T=0.693(R_1C_1+R_2C_2); \text{ symmetrical: } T\approx 1.386RC \text{ and } f\approx 0.721/(RC)$$



**Applications:** clock generator, LED flasher, pulse generator, square-wave source and timing circuits.

**മലയാളം:** Astable multivibrator-ന് stable state ഇല്ല. രണ്ട് transistors മാറിമാറി ON/OFF ആകുന്നതിനാൽ continuous square wave ലഭിക്കും.

### Design example — 1 kHz symmetrical astable

Choose  $C=0.01\ \mu\text{F}$ .  $R=0.721/(fC)=0.721/(1000\times 10\ \text{nF})=72.1\ \text{k}\Omega$ . Select  $72\ \text{k}\Omega$ . Each half-cycle is approximately  $0.693RC\approx 0.499\ \text{ms}$ , giving  $T\approx 0.998\ \text{ms}$ .

## Monostable and bistable multivibrators

### Monostable

One stable state and one temporary quasi-stable state. A trigger causes one output pulse; the circuit automatically returns after the timing interval.

Typical pulse width  $\approx 0.69RC$ , depending on circuit form.

Applications: one-shot pulse, time delay, pulse stretching and missing-pulse detection.

### Bistable

Two stable states. A trigger changes the stored state and another trigger resets it. It acts as a one-bit memory element.

**മലയാളം:** Monostable-ന് ഒരു stable state; trigger കിട്ടുമ്പോൾ ഒരു pulse ഉണ്ടാക്കി തിരിച്ചെത്തും. Bistable-ന് രണ്ട് stable states; ഒരു bit memory പോലെ പ്രവർത്തിക്കുന്നു.

## M4.04

2 hours • Applying

## Schmitt trigger

A Schmitt trigger is a regenerative comparator with two switching thresholds. It changes state at the upper trigger point (UTP) for a rising input and at the lower trigger point (LTP) for a falling input.

Hysteresis voltage  $V_H = V_{UTP} - V_{LTP}$

Hysteresis prevents repeated switching due to noise near a single threshold. Applications include waveform shaping, noisy-signal restoration, level detection and switch debouncing.

**മലയാളം:** Schmitt trigger-ന് UTP, LTP എന്ന രണ്ട് threshold ഉണ്ട്. അവയുടെ വ്യത്യാസമാണ് hysteresis voltage. Noise ഉള്ള slow input-നെ clean square wave ആക്കാൻ ഉപയോഗിക്കുന്നു.

### Simple threshold example

If  $UTP = 3\text{ V}$  and  $LTP = 1\text{ V}$ , hysteresis voltage  $= 3 - 1 = 2\text{ V}$ .

## M4.05

2 hours • Applying

## UJT relaxation oscillator

A capacitor charges through  $R$  toward  $V_{BB}$ . When its voltage reaches the UJT peak voltage, the UJT turns on and rapidly discharges the capacitor. When current falls below the valley current, the UJT switches off and charging starts again. Capacitor voltage is sawtooth-like; the output pulse at a base terminal is sharp.

$V_P \approx \eta V_{BB} + V_D$ ;  $T \approx RC \ln[1/(1-\eta)]$  (neglecting diode term);  $f \approx 1/\{RC \ln[1/(1-\eta)]\}$

**മലയാളം:** Capacitor  $R$  വഴി charge ചെയ്യും. Voltage UJT peak voltage എത്തുമ്പോൾ UJT ON ആയി capacitor വേഗത്തിൽ discharge ചെയ്യും. ഇതിലൂടെ sawtooth waveform, sharp pulse എന്നിവ ലഭിക്കും.

### Design example — approximately 1 kHz

Let  $\eta = 0.6$  and  $C = 0.1\text{ }\mu\text{F}$ .  $R = 1/[fC \ln(1/(1-\eta))] = 1/[1000 \times 0.1\text{ }\mu\text{F} \times \ln(2.5)] = 10.9\text{ k}\Omega$ . Use about  $11\text{ k}\Omega$  and verify against actual UJT parameters.

**Applications:** ramp generation, triggering SCRs/triacs, timing and simple sweep circuits.

**Series Test II focus:** cutoff and saturation, astable design, monostable/bistable operation, Schmitt-trigger thresholds and UJT relaxation-oscillator frequency.

FORMULA BANK

# Essential equations and design sequence

## Bias and CE amplifier

$$V_B = V_{CC} R_2 / (R_1 + R_2)$$
$$V_E \approx V_B - 0.7$$
$$I_E \approx V_E / R_E$$
$$r'_e \approx 25 \text{ mV} / I_E$$
$$A_v \approx -R'_C / r'_e$$

## Multistage and bandwidth

$$A_{\text{total}} = A_1 A_2 \dots$$
$$G_{v,\text{dB}} = 20 \log |A_v|$$
$$BW = f_H - f_L$$

## Tuned circuit

$$f_r = 1 / (2\pi \sqrt{LC})$$
$$Q = f_r / BW$$
$$BW = f_r / Q$$

## Feedback

$$A_f = A / (1 + A\beta)$$

$$\text{Desensitivity} = 1 + A\beta$$

$$A_{f,+} = A / (1 - A\beta)$$

## Oscillators

$$\text{RC phase shift: } f = 1 / (2\pi RC\sqrt{6})$$

$$\text{Wien: } f = 1 / (2\pi RC)$$

$$\text{Hartley: } f = 1 / (2\pi\sqrt{L C_T})$$

$$\text{Colpitts: } C_{eq} = C_1 C_2 / (C_1 + C_2)$$

## Pulse circuits

$$\text{Astable: } T = 0.693(R_1 C_1 + R_2 C_2)$$

$$\text{Schmitt: } V_H = U_{TP} - L_{TP}$$

$$\text{UJT: } f \approx 1 / [RC \ln(1 / (1 - \eta))]$$

**Design habit:** write the target frequency or gain, choose one practical component value, calculate the remaining value, select a standard value, and finally verify the achieved result and component ratings.

## FINAL REVISION

# One-page memory map

## Module 1

- Bias and Q-point
- Load line
- CE parameters
- Emitter follower
- RC / transformer / direct coupling
- Frequency response

## Module 2

- Series and parallel resonance
- $f_r$ , Q and BW
- Single / double tuned
- Classes A, B, AB, C
- Push-pull and efficiency
- Crossover distortion

## Module 3

- $A_f = A/(1+A\beta)$
- Four feedback topologies
- Negative-feedback benefits
- Barkhausen criterion
- RC, Wien, Hartley, Colpitts
- Crystal oscillator

## Module 4

- Cutoff and saturation
- Astable design
- Monostable / bistable
- UTP, LTP and hysteresis
- UJT charging/discharging
- Pulse applications

**Malayalam last-day cue:** Bias → gain → coupling → resonance → power classes → feedback → Barkhausen → oscillators → switch → multivibrators → Schmitt → UJT എന്ന ക്രമത്തിൽ ആവർത്തിക്കുക.

## Likely 1-mark areas

- Define Q-point.
- State emitter-follower voltage gain.
- Write resonant-frequency formula.
- State maximum Class-B efficiency.
- State Barkhausen criterion.
- Define hysteresis voltage.

## Likely short answers

- Fixed versus divider bias.
- RC versus transformer coupling.



- Single versus double tuned.
- Class A/B/AB/C comparison.
- Negative-feedback advantages.
- Monostable versus bistable.

## Likely design questions

- CE Q-point and voltage gain.
- Overall multistage gain in dB.
- LC resonance and bandwidth.
- RC phase-shift / Wien oscillator.
- Astable multivibrator.
- UJT relaxation oscillator.

### PRACTICE MODEL QUESTION PAPER

## ELECTRONIC CIRCUITS • Maximum Marks: 75 • Time: 3 Hours

This is a fresh practice model prepared from the supplied syllabus and is not an official board question paper.

### PART A — Answer all questions. Each carries 1 mark. [5 × 1 = 5]

1. What is the purpose of transistor biasing?
2. Write the relation between resonant frequency, Q and bandwidth.
3. State the maximum theoretical efficiency of a Class-B push-pull amplifier.
4. State the Barkhausen criterion.
5. Define hysteresis voltage of a Schmitt trigger.

### PART B — Answer any five. Each carries 8 marks. [5 × 8 = 40]

1. Explain voltage-divider bias and calculate the Q-point for  $V_{CC}=12\text{ V}$ ,  $R_1=47\text{ k}\Omega$ ,  $R_2=10\text{ k}\Omega$ ,  $R_C=2.2\text{ k}\Omega$  and  $R_E=1\text{ k}\Omega$ .
2. Explain the CE amplifier, its gains, impedances and frequency response.
3. Compare RC, transformer and direct coupling. Two stages have gains 20 and 30; find total gain and dB gain.
4. Explain single- and double-tuned amplifiers. A circuit has  $L=100\text{ }\mu\text{H}$ ,  $C=100\text{ pF}$  and  $Q=50$ ; calculate  $f_r$  and bandwidth.
5. Compare Classes A, B, AB and C and explain crossover distortion.
6. Explain negative feedback and the four feedback topologies. Find  $A_f$  for  $A=100$  and  $\beta=0.04$ .
7. Explain astable, monostable and bistable multivibrators with applications.

### PART C — Answer any two. Each carries 15 marks. [2 × 15 = 30]

1. Explain transistor biasing, single-stage CE amplification and multistage coupling. Include design equations and frequency response.

2. Explain RC phase-shift, Wien bridge, Hartley, Colpitts and crystal oscillators. Design an RC phase-shift oscillator for 1 kHz using  $C=0.01\ \mu\text{F}$ .
3. Explain transistor switching, BJT astable multivibrator, Schmitt trigger and UJT relaxation oscillator. Design a 1 kHz symmetrical astable using  $C=0.01\ \mu\text{F}$ .

Course 2013

## COMPLETE ANSWER KEY

# Model-paper answers

## Part A answers

| No. | Answer                                                                                 |
|-----|----------------------------------------------------------------------------------------|
| 1   | To establish and stabilise a suitable DC operating point in the active region.         |
| 2   | $Q=f_r/BW$ , therefore $BW=f_r/Q$ .                                                    |
| 3   | 78.5%.                                                                                 |
| 4   | Loop-gain magnitude $ A\beta =1$ and total loop phase shift $0^\circ$ or $360^\circ$ . |
| 5   | $V_H=V_{UTP}-V_{LTP}$ .                                                                |

## Part B full answers

### B1. Voltage-divider bias and Q-point

Voltage-divider bias uses  $R_1$  and  $R_2$  to establish a nearly fixed base voltage.  $R_E$  introduces negative DC feedback and improves thermal and beta stability.

$$\begin{aligned} V_B &= V_{CC} R_2/(R_1+R_2) \\ &= 12 \times 10/(47+10) \\ &= 2.105\text{ V} \end{aligned}$$

$$\begin{aligned} V_E &= V_B - 0.7 = 1.405\text{ V} \\ I_E &= V_E/R_E = 1.405\text{ mA} \\ I_C &\approx I_E = 1.405\text{ mA} \end{aligned}$$

$$\begin{aligned} V_C &= V_{CC} - I_C R_C \\ &= 12 - (1.405\text{ mA} \times 2.2\text{ k}\Omega) \\ &= 8.909\text{ V} \end{aligned}$$

$$V_{CE} = V_C - V_E = 8.909 - 1.405 = 7.504\text{ V}$$

$$Q\text{-point} \approx (V_{CE}=7.50\text{ V}, I_C=1.41\text{ mA}).$$

The emitter resistor opposes a temperature-induced rise of current: increased current raises  $V_E$ , reduces  $V_{BE}$  and limits the original rise.

## B2. CE amplifier

1. Input is applied to base–emitter and output is taken from collector–emitter.
2. Voltage-divider resistors establish the Q-point.
3. Coupling capacitors pass AC but block DC.
4. The emitter bypass capacitor increases midband gain.
5. Output is approximately 180° out of phase with input.

Important relations:

$$r_e' \approx 25 \text{ mV}/I_E$$

$$A_v \approx -(R_C || R_L)/r_e'$$

$$A_i \approx \beta$$

$$A_p = |A_v|A_i$$

$$Z_{in} \approx R_1 || R_2 || r_{\pi}$$

$$Z_{out} \approx R_C$$

$$BW = f_H - f_L$$

Frequency response:

- Low frequency: capacitor reactance reduces gain.
- Midband: gain is approximately constant.
- High frequency: transistor and stray capacitances reduce gain.

## B3. Coupling methods and overall gain

RC coupling:

- Good audio response, economical and compact.
- Poor impedance matching and reduced gain at extremes.

Transformer coupling:

- Good impedance matching and power transfer.
- Bulky, costly and restricted frequency response.

Direct coupling:

- Passes DC and very low frequency.
- Susceptible to drift and bias interaction.

$$\text{Overall voltage gain} = 20 \times 30 = 600.$$

$$\text{Gain in dB} = 20 \log_{10}(600) = 55.56 \text{ dB}.$$

## B4. Tuned amplifiers and numerical answer

A single-tuned amplifier uses one parallel LC collector load and gives maximum gain near resonance. It is simple but has a gain–bandwidth limitation.

A double-tuned amplifier uses two coupled resonant circuits. Coupling controls the response: under-coupled gives a narrow single peak, critical coupling gives a broader useful response, and over-coupled gives two peaks.

Given  $L=100\text{ }\mu\text{H}$  and  $C=100\text{ pF}$ :

$$f_r = 1/(2\pi\sqrt{LC}) \\ = 1.59\text{ MHz}$$

$$Q = f_r/BW$$

$$BW = f_r/Q = 1.59\text{ MHz}/50 = 31.8\text{ kHz}.$$

## B5. Power-amplifier classes

Class A: conducts  $360^\circ$ , excellent linearity, low efficiency; maximum 25% with resistive load and 50% with transformer coupling.

Class B: conducts  $180^\circ$ . Two devices in push-pull reproduce the full cycle. Maximum theoretical efficiency is 78.5%.

Class AB: conducts more than  $180^\circ$  but less than  $360^\circ$ . A small standing current reduces crossover distortion.

Class C: conducts less than  $180^\circ$ . It is highly nonlinear but efficient and is used with an LC tuned load in RF circuits.

Crossover distortion occurs around the zero crossing when neither Class-B transistor conducts because the input has not exceeded the base-emitter threshold.

## B6. Negative feedback and topologies

Negative feedback returns a fraction of output in opposition to the input.

$$A_f = A/(1+A\beta)$$

Benefits:

- stable gain
- lower distortion and noise
- wider bandwidth
- controllable input and output impedances

Topologies:

1. Voltage-series:  $Z_{in}$  increases,  $Z_{out}$  decreases.
2. Voltage-shunt:  $Z_{in}$  decreases,  $Z_{out}$  decreases.
3. Current-series:  $Z_{in}$  increases,  $Z_{out}$  increases.
4. Current-shunt:  $Z_{in}$  decreases,  $Z_{out}$  increases.

For  $A=100$  and  $\beta=0.04$ :

$$A\beta=4$$

$$A_f=100/(1+4)=20.$$

## B7: Multivibrators

Astable:

- no stable state
- switches continuously
- produces rectangular wave
- used in clocks and pulse generators

Monostable:

- one stable and one quasi-stable state
- trigger produces one pulse and circuit returns automatically
- used in timers, pulse stretching and delay circuits

Bistable:

- two stable states
- set/reset triggers select the state
- used as a flip-flop, one-bit memory and counter element.

## C1. Biasing, CE amplifier and multistage amplifier

### Introduction:

An amplifier needs a stable DC Q-point and an AC signal path. Biasing fixes the DC conditions; coupling and bypass components determine AC behaviour.

### A. Biasing

- Need: active-region operation and undistorted signal swing.
- DC load line:  $V_{CE} = V_{CC} - I_{C} R_{C}$ .
- Q-point is the no-signal operating point.
- Fixed bias is simple but beta dependent.
- Voltage-divider bias gives better stability through a nearly fixed  $V_B$  and emitter feedback.

### Approximate equations:

$$V_B = V_{CC} \frac{R_2}{R_1 + R_2}$$

$$V_E = V_B - 0.7$$

$$I_E = V_E / R_E$$

$$V_{CE} = V_{CC} - I_{C} R_{C} - I_{E} R_E$$

### B. CE amplifier

- input at base, output at collector
- high voltage and power gain
- 180° phase reversal
- coupling capacitors block DC
- emitter bypass capacitor raises AC gain

$$r_e' \approx 25 \text{ mV} / I_E$$

$$A_v \approx -(R_C || R_L) / r_e'$$

$$A_i \approx \beta$$

$$A_p = |A_v| A_i$$

### C. Frequency response

- low frequency: coupling/bypass effects
- midband: almost constant gain
- high frequency: junction/stray capacitance
- $BW = f_H - f_L$

### D. Multistage amplifier

Need: gain greater than one stage can supply.

$A_{total} = A_1 A_2 \dots$ ; total dB gain is the sum of stage dB gains.

### Coupling:

- RC for audio voltage amplification
- transformer for impedance matching and power transfer
- direct for DC and very low frequency

### Conclusion:

Correct biasing, suitable coupling and controlled frequency response are jointly required for a practical amplifier.

## C2: Sinusoidal oscillators and RC design

Oscillator principle:

An oscillator uses positive feedback and a frequency-selective network to convert DC power to AC.

Barkhausen criterion:

$|A\beta|=1$  and loop phase= $0^\circ/360^\circ$ .

RC phase-shift:

- transistor amplifier gives  $180^\circ$
- three RC sections give another  $180^\circ$
- $f=1/(2\pi RC\sqrt{6})$

Wien bridge:

- lead-lag RC bridge
- low-distortion audio output
- $f=1/(2\pi RC)$
- amplifier gain condition approximately 3

Hartley:

- inductive divider L1 and L2
- $f=1/[2\pi\sqrt{C(L1+L2+2M)}]$

Colpitts:

- capacitive divider C1 and C2
- $C_{eq}=C1C2/(C1+C2)$
- $f=1/(2\pi\sqrt{LC_{eq}})$

Crystal:

- uses piezoelectric resonance
- extremely high Q and excellent frequency stability
- used for clocks and communication frequency standards

Design for RC phase-shift oscillator:

$f=1$  kHz,  $C=0.01$   $\mu$ F= $10$  nF

$R=1/(2\pi fC\sqrt{6})$

$=6.50$  k $\Omega$

Choose approximately  $6.49$  k $\Omega$  or a nearby practical value and verify actual frequency.

### C3: Switching and pulse circuits

#### A. Transistor switch

- cutoff:  $I_B \approx 0$ ,  $I_C \approx 0$ , open switch
- saturation: sufficient base current,  $V_{CE(sat)}$  small, closed switch
- $R_B = (V_{drive} - V_{BE}) / I_B$

#### B. Astable multivibrator

- no stable state
- cross-coupled transistor stages switch alternately
- complementary rectangular outputs
- $T = 0.693(R_1C_1 + R_2C_2)$
- symmetrical  $f \approx 0.721 / (RC)$

Design:

$f = 1 \text{ kHz}$ ,  $C = 0.01 \mu\text{F}$

$R = 0.721 / (fC) = 72.1 \text{ k}\Omega$

Choose  $72 \text{ k}\Omega$ .

#### C. Schmitt trigger

- regenerative comparator
- switches at UTP for rising input and LTP for falling input
- hysteresis  $V_H = UTP - LTP$
- converts a noisy/slow waveform into a clean rectangular waveform

#### D. UJT relaxation oscillator

- capacitor charges through R
- UJT turns on at peak voltage and rapidly discharges capacitor
- UJT switches off near valley condition
- capacitor voltage is sawtooth-like
- $f \approx 1 / [RC \ln(1 / (1 - \eta))]$

Applications:

pulse generation, timing, waveform shaping, ramp generation and SCR/triac triggering.

### REFERENCES FROM THE SYLLABUS



# Textbooks and online resources

Course 3043

| Code | Book / Author                                                                                        |
|------|------------------------------------------------------------------------------------------------------|
| T1   | R. S. Sedha, <i>A Text Book of Applied Electronics</i> , S. Chand.                                   |
| R1   | N. N. Bhargava, Kulshreshtha and S. C. Gupta, <i>Basic Electronics and Linear Circuits</i> , TMH.    |
| R2   | Robert Boylestad, <i>Electronic Devices and Circuits</i> , PHI.                                      |
| R3   | Anil K. Maini and Varsha Agarwal, <i>Electronic Devices and Circuits</i> , Wiley India.              |
| R4   | David A. Bell, <i>Electronic Devices and Circuits</i> , PHI.                                         |
| R5   | Allen Mottershead, <i>Electronic Devices and Circuits: An Introduction</i> , Prentice-Hall of India. |

**Online resources listed in the syllabus:** Electronics Tutorials, ElProCus, BrainKart Electronics Engineering and Electrical4U.